

Reaction Scheme — Quick Guide

PAAF · define reactions with mapped SMILES, export typed structures

Where it is

Open the **Builder** page and select the **Reaction scheme** tab at the top. It has three steps of its own: **1 · Scheme** (describe the chemistry), **2 · Force field** (how to type it), **3 · Export** (write the files). It is different from the **Reactions** tool in the sidebar: the scheme *defines* reactions from SMILES and exports reference structures; the Reactions tool *learns* a transformation from uploaded reactant/product structure files and can apply it to a packed cell (crosslinking).

Atom maps in 60 seconds

A map number tags one atom so PAAF can follow it through the reaction. Write it in brackets after the element: `[C:1]`. The same number on the reactant and product side means *the same atom*.

Rule 1 Map only the atoms whose bonds change.

Rule 2 Every number must be unique within one side.

Rule 3 A number present in the reactants but missing from the products is a leaving group (it is deleted).

Rule 4 Numbers are per reaction — Reaction 2 may reuse :1.

Rule 5 Hydrogens are completed automatically: `[C:2]` keeps its normal hydrogens (it is a tag, not a valence statement). An explicit count like `[CH2:2]` is respected as written.

Example 1 · Esterification (the simplest case)

Acetic acid + ethanol → methyl-ester + water. The acid's OH oxygen (:1) leaves as part of water; the alcohol carbon (:2) ends up bonded to the ester oxygen.

Reactant 1	<code>CC(=O)[OH:1]</code>
Reactant 2	<code>[C:2]CO</code>
Product 1	<code>CC(=O)O[C:2]</code>
Product 2	<code>O</code>

:1 appears only on the reactant side → leaving group (Rule 3). :2 appears on both sides → the bond from that carbon to the ester oxygen is the bond being formed.

Example 2 · Epoxide ring-opening (regioselective)

A model for ENR + maleic acid: the acid oxygen attacks the **tertiary** epoxide carbon; the tertiary C–O(epoxide) bond breaks; the epoxide oxygen stays on the **secondary** carbon and becomes an OH. Regiochemistry is stated simply by drawing which mapped carbon receives the new bond.

Epoxide	<code>C[C:1]1(C)[O:2][CH:3]1C</code>
Maleic acid	<code>O=C(O)/C=C\C(=O)[OH:4]</code>

Product	<chem>C[C:1](C)([O:4]C(=O)/C=C\C(=O)O)[CH:3](C)[O:2]</chem>
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:1 = tertiary carbon (no H, three C neighbours) · :2 = epoxide oxygen · :3 = secondary carbon · :4 = attacking acid oxygen. In the product, :4 is bonded to :1 (bond formed), :2 is no longer bonded to :1 (bond broken) and hangs off :3 as a hydroxyl — its H arrives by automatic hydrogen completion (Rule 5).

Example 3 · Condensation with a leaving group

Two carboxylic acids → anhydride + water. One acid keeps its oxygen as the bridge (: 6); the other's OH (: 5) is not in any product, so it leaves.

Reactant 1	<chem>CC(=O)[OH:5]</chem>
Reactant 2	<chem>CC(=O)[OH:6]</chem>
Product 1	<chem>CC(=O)[O:6]C(C)=O</chem>
Product 2	<chem>O</chem>

Example 4 · A full bridge: ENR–maleic acid–ENR (model)

Both COOH groups of maleic acid open one epoxide each — two bond edits in a single reaction. Three reactant molecules, one product molecule. This is the small-molecule model of an ENR/MAH crosslink; the same pattern with a COOH-terminated PBS instead of the second epoxide gives the ENR–MAH–PBS graft.

Epoxide A	<chem>C[C:1]1(C)[O:2][CH:3]1C</chem>
Maleic acid	<chem>[OH:4]C(=O)/C=C\C(=O)[OH:5]</chem>
Epoxide B	<chem>C[C:6]1(C)[O:7][CH:8]1C</chem>
Product	<chem>C[C:1](C)([O:4]C(=O)/C=C\C(=O)[O:5][C:6](C)(C)[CH:8](C)[O:7])[CH:3](C)[O:2]</chem>

Every example on these pages was validated against PAAF's own reaction parser before being printed here — paste them verbatim to try the tool.

Step 2 · Force field

Pick the force field (e.g. OPLS 2005 via DL_FIELD). The DL_FIELD lib dir is inherited from the pipeline automatically; fill it only if you want a different installation. Keep *Energy-minimise each structure before typing* ticked, and tick *Run dl_field to produce LAMMPS .data* for typed output.

Step 3 · Export

Choose the reactions folder and press **Export .data....** Each reaction gets its own folder containing:

<chem>reactant.xyz</chem> / <chem>product.xyz</chem>	3D structures of each side
<chem>reactant.mol2</chem> / <chem>product.mol2</chem>	same, with bond orders
<chem>reactant.data</chem> / <chem>product.data</chem>	typed LAMMPS files (if typing is on)
<chem>reaction.json</chem>	the learned template: mapped atoms, broken/created bonds, deleted atoms

If something goes wrong

A SMILES box turns red	RDKit could not parse it. Check ring closures (1...1) and brackets.
“dl_field executable not found”	Set the DL_FIELD lib dir on the Force-field step (or on the pipeline's Force-field page; the scheme inherits it).
Structures written but no .data	Open the Run log: dl_field's own message is printed there. Most often an atom it cannot type — check the structure is a real, complete molecule.
Maps look right but validation complains	The “Map numbers in use” card shows both sides; a number used twice on one side, or a product number that never appears in the reactants, is flagged.